

WHITE PAPER · SPRINT ROADMAP

MOLGANG

Road to 1,000,000 Concurrent Peers

A peer-to-peer chemistry learning game on the Knitweb — where the game *is* the protocol, and capacity grows with every player. The plan to out-scale GTA6.

Knitweb · MOLGANG

Companion to the Sprint 3–10 plan · 2026.06

github.com/Knitweb/molgang · knitweb.github.io/molgang

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01 · EXECUTIVE SUMMARY

| Executive Summary

MOLGANG is a peer-to-peer chemistry (scheikunde) learning game in which *the game is the protocol*: every chemical bond a learner forms is a real Knit on the Knitweb ledger, every molecule is a Fiber, and classmates validate the chemistry by voting with PLS pulses settled by a real Byzantine-fault-tolerant Proof-of-Useful-Work quorum. The goal of this roadmap is audacious and concrete: **one million concurrent peers** — out-scaling GTA6 — reachable precisely because on a peer fabric, players are the infrastructure.

Key Takeaways

- The edge is architectural, not budgetary. Capacity grows with players: each peer that joins also serves, so scale compounds instead of costing.
- Eight sprints, one frozen contract. Sprint 3–10 carry the project from a two-player shared world to a measured million-peer launch — ~60 new issues plus 11 gap issues (74 open), each gated and labelled.
- Security and observability gate scale. A hard security/economy/legal gate precedes mass onboarding; full instrumentation precedes the 1M test — both are sequencing guarantees, not afterthoughts.

QUESTIONS THIS DOCUMENT ANSWERS

Why can a chemistry game realistically target a million concurrent players? What are the eight sprints, and how does each move the needle toward 1M? In what order must they land, and what counts as "done"?

The Thesis & the North Star

Most multiplayer games scale by spending: more concurrent players means more servers, linearly. MOLGANG inverts that economics by running on a decentralized fabric.

The Game Is the Protocol

MOLGANG is built on the [Knitweb](#) fabric (engine: `knitweb/pulse`). The mapping is one-to-one and real — nothing is mocked:

In the game	On the Knitweb
Forming a chemical bond	a Knit — a two-party ledger transfer
Growing a molecule	a Fiber — an account-state commitment
Classmates validating chemistry	PLS pulses staked into a BFT PoUW quorum
A free start for new players	silk + pulses from the faucet

There are no NFTs. Value is pulses, reputation, and woven knowledge — useful work, not speculation.

Why One Million Is Reachable

On a server-farm game, the millionth player is a cost. On a peer fabric, the millionth player is a [relay](#), a [validator](#), and a [teacher](#) — capacity the network gains, not capacity it must buy.

The engineering task is therefore not "afford a million players" but "let a peer fabric self-organize, stay honest, and stay fast as it grows." That is exactly what Sprints 4–10 build.

Where We Stand

Dimension	Today (Sprint 1–2)	Target (Sprint 10)
Live nodes	One always-on node (5mart.ml) + relay	Multi-region reconciling relay mesh
Concurrent players	Two-player shared world	1,000,000 concurrent peers
Peer discovery	Hand-typed / relay-brokered	Kademlia DHT + NAT hole-punch
Content	Core chemistry + multilingual KG	Graded curriculum, quests, seasons

03 · THE ARC

| The Ten-Sprint Arc

Sprint 1–2 are done or in progress. This roadmap fills Sprint 3–10 — short, dependency-ordered increments (no fixed calendar dates). Each line states the sprint's theme and how it advances the million-peer goal.

Sprint	Theme — and how it advances 1M
1 · done	Foundation — UI, deploy to 5mart.ml, CI, monitor tab, p2p relay, PoUW accounting. <i>A live single node.</i>
2 · wip	Shared world + Django dapp + multilingual KG. <i>Two-player real-time and i18n content.</i>
3	Shared-world + multilingual KG + one canonical /api engine — converge the Python bar, Django dapp and PHP node on one frozen versioned /api ; Channels push (not poll); presence/reconnect + co-weaving spirals; trilingual EN/RU/ZH(+AR) knowledge graph. <i>One wire protocol and a push transport for any peer.</i>
4	P2P hardening & peer discovery — Kademlia DHT routing, STUN NAT hole-punch, equivocation quarantine + partial Merkle proofs, FabricNode/AsyncioP2PNode unification, fiber-tension v1. <i>A self-organising, Byzantine-resistant, many-thousand-peer mesh.</i>
5	Relay mesh & horizontal scale — multi-relay client with failover, relay-to-relay anti-entropy gossip, concept-sharded knowledge graph, region-aware bootstrap, durable storage. <i>One relay becomes a reconciling mesh across regions.</i>
6	PoUW economy, tokenomics & security gate — fix the certificate key-leak, faucet/sybil + rate-limit + key hygiene, collateral/snap-slash, rewards for relay operators, identity recovery, content moderation, COPPA/GDPR-K. <i>Sybil-resistant and safe BEFORE mass onboarding.</i>
7	Content depth — graded chemistry curriculum + reactions-with-conditions, machine-woven concept graph, quests/achievements/seasons, reputation ladder, translation pipeline. <i>Retention turns capacity into sustained concurrency.</i>
8	Client, onboarding & accessibility — installable offline-first mobile PWA, EN/NL i18n, first-run tutorial, a11y/RTL, PHP-UI parity, edge/CDN app-shell. <i>Low-friction join from any phone, network, or language.</i>
9	Load testing, bot swarms & observability — agentplay LLM bot swarm, ramp-to-N-thousand load harness, Prometheus/Grafana + distributed tracing, SLO budgets, chaos, autoscale. <i>Make 1,000,000 a measured engineering target.</i>
10	Launch — 1M concurrent & the GTA6 benchmark — multi-region relay fleet, flash-crowd faucet hardening, the measurement standard, the million-peer run, a public GTA6-comparison scoreboard, runbook + cost model. <i>Demonstrate it.</i>

Across Sprint 3–10 the plan files ~60 new issues plus 11 gap issues (74 open), every one gated and labelled by area, type, priority, and MoSCoW. The live GitHub Pages roadmap renders this board straight from the Issues API, so progress is always current.

Execution: Ordering, Gates & What "Done" Means

Sequence is a feature. Four orderings are guaranteed because each later sprint depends on an earlier one being true.

Guaranteed Orderings

1. Contract before clients — Sprint 3 freezes one /api before any client, relay, or bot is built on it.
2. Discovery before mesh — Sprint 4's DHT lands before Sprint 5 fans peers across relays.
3. Security before mass onboarding — Sprint 6's economy/safety gate precedes the PWA (8) and launch (10).
4. Observability before the test — Sprint 9's metrics, tracing, and SLOs precede the million-peer run (10).

What "Done" Means

Sprint 10 defines the measurement standard explicitly — what "concurrent" counts, how it is observed, and what threshold beats GTA6 — then runs the test (agentplay swarms plus real peers) and publishes the methodology, results, and a public scoreboard. A claim of one million peers is only true once it is [measured under a published standard](#), not asserted.

CALL TO ACTION

Open knitweb.github.io/molgang to watch the live board, or github.com/Knitweb/molgang/milestones to pick up a Sprint 3 issue. Start where the contract starts: one frozen /api .

| Appendix — Vocabulary & Links

Knitweb Vocabulary

Web · Knitweb — the peer-to-peer fabric the game runs on.

Knit — a chemical bond, realised as a two-party ledger transfer.

Fiber — a molecule, realised as an account-state commitment.

Pulse / PLS — the validation vote and pay-token; staked into the PoUW quorum.

Silk — the free starting allowance new players draw from the faucet.

Spider — a worker/relay operator that serves the fabric. NEVER "LOOM"

PoUW — Proof of Useful Work: the BFT quorum that settles whether woven chemistry is correct.

Links

- Live roadmap — knitweb.github.io/molgang
- Milestones & issues — github.com/Knitweb/molgang/milestones
- Engine — github.com/knitweb/pulse

Acknowledgements

This roadmap is the companion document to the approved Sprint 3–10 plan. It is meant to be read in ten minutes and shared widely — the clearest possible statement of how a chemistry game intends to out-scale GTA6 by making every learner part of the network.